Entity-Oriented Knowledge Discovery in Knowledge-Intensive Domains - Third Year PhD Report

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1 Research Activity

Motivation from SOA

In my research, I focus on the extraction of knowledge from knowledge-intensive domains and on methods to allow end users to access this knowledge. In domains such as legal or investigative, entities are crucial [1]: legal judgments contain several mentions of persons, and investigators are often interested in finding mentions of persons or locations, e.g., from instant messaging data obtained from mobile phones. Therefore, extracting entities and organizing them, for example in a knowledge graph [2; 3; 4], can simplify the exploration and interpretation activities of users (investigators, lawyers, or judges). Nevertheless, user interfaces (UIs) paired with search engines and question answering systems are vital to accomplish the users' need of knowledge discovery. Research work, indeed, confirms this interest developing UIs empowered by search engines with entity-based faceted search [3; 4] or question answering [5; 6].

The research activities I conducted during these years have been directed towards the extraction and organization of entity-based knowledge (e.g., persons, locations, or organizations mentioned in the documents) [7; 8; 9; 10; 11], and its utilization by end users, including the design of frameworks that provide access to faceted search and question answering through user interfaces [12; 11; 10] and an efficient question-answering approach [13].

Definitions

Entity extraction is defined as the task that comprehends identifying mentions of entities from text (named entity recognition – NER) and linking them to an item of a knowledge base (named entity linking – NEL). However, specific domains like the legal one, often mention entities not present in public general-purpose knowledge bases such as Wikidata. The identification of new entities can be addressed by the two additional subtasks: NIL prediction detects which mentions refer to unknown entities (that are not-in-lexicon) and NIL clustering groups together the mentions referring to the same new entity [8]. These tasks were underrepresented in the literature (only a minority of studies tackled NIL entities in 2022 [14]) and have been addressed in my research by adapting an existing dataset to an incremental NIL-aware scenario and providing a baseline extraction pipeline [8] and by the creation of a NIL-aware datasets for entity linking on tabular data [9].

NIL

Domain specific

Another important issue of specific domains is the limited availability of labeled data for training neural models [15], especially in sensitive domains where documents are protected for privacy-related reasons [16]. Focusing the legal domain, in [11] several legal judgments in Italian were annotated highlighting the challenges of using general-domain models for entity extraction in this domain. Similarly, in [10] we annotated some WhatsApp chats in Italian from suspects' mobile phones highlithing the same issue. Indeed, in both cases the language is different: judgments usually follow a structure (i.e., they can be divided in different sections), use a formal language and have peculiarities (e.g., names in capital letters) for which general-purpose models may not be familiar with; WhatsApp chats, on the other hand, are particularly informal and often contain

typos and abbreviations.

Subsequently, different adaptation techniques (in-domain masked language modeling and in-domain task-oriented fine-tuning, in different combinations) have been explored with the annotated judgments evidencing that in-domain task-specific fine-tuning (i.e., fine-tuning a BERT model for NER on in-domain data) is a good compromise between performance and computational costs, while also performing in-domain masked language modeling achieves slightly better performance but at higher computational costs [7].

UIs and organization

With respect to organizing and accessing the extracted knowledge, I contributed to the development of an entity-centric infrastructure to annotate and manage court judgments [11]. Similarly in [10] we worked on WhatsApp chats also representing chat-based relations in a knowledge graph (e.g., for visualizing the number of messages shared between the suspect and the contacts). Both infrastructures provide users with human-in-the-loop [17] functionalities to correct wrong annotations and entity-based exploration tools, including faceted search.

SOA

Finally, with respect to question answering on legal documents, users that prefer to avoid the use of APIs with private data are forced to rely on smaller open-weight models, running on local hardware with stricter resource limits. For this reason, we worked on a novel and efficient approach for question answering on external knowledge (where with external I mean knowledge that has not been fed to the model during its training phase) named ReFactX [13]. While existing approaches like RAG [18; 19] and tool-use [20; 21; 22] are able to perform this task, they depend on external tools or pipelines suffering from error-propagation and requiring the model to parse tool outputs, increasing the number of tokens to process. Alternatives include custom model architectures built specifically for accessing external knowledge [23; 24]; however, these architectural modifications are not compatible with top-tier powerful language models complicating their usability.

ReFactX

ReFactX, instead, uses constrained generation, a technique that constrains a generative model (in our case by limiting its vocabulary) to generate controlled outputs, such as entity titles for performing entity linking [25], or to follow formal grammars for code generation and other NLP tasks [26]. With constrained generation and a prefix-tree offloaded to disk, ReFactX is able to access external facts efficiently (requiring only +1.3% generation time) and it can scale to big knowledge bases (we were able to represent 800M facts from Wikidata using 95GB on disk), achieving competitive results on benchmark datasets. The following example depicts ReFactX answering a question generating two facts from the knowledge base.

```
Question: When was the director of Slumdog Millionaire born?
    I need to find who is the director and when he was born.
    Fact: <Slumdog Millionaire> <director> <Danny Boyle> .
    Fact: <Danny Boyle> <date of birth> <1956-10-20> .
Answer: October 20, 1956
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Normally a generative language models, given a prefix, iteratively predicts one-token at a time selecting the token that is most likely (in case of greedy decoding) to follow the prefix from the entire vocabulary. With constrained generation, instead, we limit the vocabulary to the allowed tokens: given the prefix "<Danny Boyle> <" the allowed continuations are "date of birth>..." or the other existing facts such as "given name>...". In this way, with constrained generation and a prefix tree offloaded to disk, ReFactX is able to generate facts from a knowledge base without additional search engines or embedding models.

2 Ph.D Progress

2.1 Interdisciplinary Courses

I obtained 3 out of 3 interdisciplinary CFUs.

1. Statistics with R II (1 CFU)
2. Reading the world through gendered lens (2 CFUs)

2.2 Disciplinary Courses

I obtained 8 out of 8 disciplinary CFUs.

1. Image Acquisition Pipelines (2 CFUs)
2. Causal Network: Learning and Inference (2 CFUs)
3. Mediterranean Machine Learning Summer School (2 CFUs)
4. Graph Theory and Algorithms (2 CFUs)

2.3 Period Abroad

I completed my 6 months abroad visiting Sahar Vahdati's Research Group at TU Dresden / InfAI[1]. During this period I worked on ReFactX and our collaboration is still ongoing.

2.4 Co-authored Publications

The following list also includes two papers in the scope of my PhD that I coauthored and that have been submitted and presented before the official start of my PhD (1st November 2022), and published in 2023.

1. [13] Riccardo Pozzi, Matteo Palmonari, Andrea Coletta, Luigi Bellomarini, Jens Lehmann, Sahar Vahdati. ReFactX: Scalable Reasoning with Reliable Facts via Constrained Generation. Submitted to the ISWC 2025 Research Track, 2025.
2. [12] Ruben Agazzi, Renzo Alva Principe, Riccardo Pozzi, Marco Ripamonti, Matteo Palmonari. DAVE: A Framework for Assisted Analysis of Document Collections in Knowledge-Intensive Domains. In Proceedings of the IJCAI 2025 Demonstration Track, 2025, (Accepted for presentation).
3. [10] Riccardo Pozzi, Valentina Barbera, Renzo Alva Principe, Davide Giardini, Riccardo Rubini, Matteo Palmonari (2025). Combining Knowledge Graphs and NLP to Analyze Instant Messaging Data in Criminal Investigations. In International Conference on Web Information Systems Engineering (pp. 427-442). Springer, Singapore.
4. [11] Valerio Bellandi, Christian Bernasconi, Fausto Lodi, Matteo Palmonari, Riccardo Pozzi, Marco Ripamonti, Stefano Siccardi (2024). (All authors share the same contribution). An entity-centric approach to manage court judgments based on Natural Language Processing. Computer Law & Security Review, 52, 105904.
5. [7] Riccardo Pozzi, Riccardo Rubini, Christian Bernasconi, and Matteo Palmonari (2023, November). Named Entity Recognition and Linking for Entity Extraction from Italian Civil Judgements. In International Conference of the Italian Association for Artificial Intelligence (pp. 187-201). Cham: Springer Nature Switzerland.

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6. [8] Riccardo Pozzi, Federico Moiraghi, Fausto Lodi, and Matteo Palmonari. 2023. Evaluation of Incremental Entity Extraction with Background Knowledge and Entity Linking. In Proceedings of the 11th International Joint Conference on Knowledge Graphs (IJCKG '22). Association for Computing Machinery, New York, NY, USA, 30–38.

   I presented this paper remotely at the end of October 2022.

7. [9] Mattia Marzocchi, Marco Cremaschi, Riccardo Pozzi, Roberto Avogadro, and Matteo Palmonari. 2023. MammoTab: a giant and comprehensive dataset for Semantic Table Interpretation. In CEUR WORKSHOP PROCEEDINGS (Vol. 3320, pp. 28-33). CEUR-WS.

   This work has been presented at the end of October 2022 and is part of the thesis of a master's student in Data Science for whom I've been the thesis co-relator.

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